Certified Low-Rank BSD Leading Terms and Integral Arithmetic Boundaries

A Ram-Calculus Study of M002

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M002

October 8, 2026

Abstract

We study the pinned manuscript *Exact Birch–Swinnerton-Dyer Formula from Low Selmer Corank*, keeping its integral arithmetic arguments distinct from finite analytic certificates. The whole source structure is mapped, but only explicitly logged passages are literally read. Actual private Mellin, panel, regulator and saturation proofs supply the transfer route. For ranks zero and one we derive direct scalar kernels for the uncompleted leading term, prove positive rational bounds and a refinable weighted-panel certificate for the rank-one kernel, and obtain a strictly smaller coefficient-tail majorant than the inherited log-atom bound. We retain the source's total real period, height diagonal, full lattice, torsion factors and squared indices. A defect identity exposes the exact integral-unit input, while a conditional odd-integer interval criterion works with a proved upper bound on the point index. These are local analytic and logical extensions, not a proof of low-Selmer converses, theta extraction, integral BSD, or the exact order of the Tate–Shafarevich group. Contribution assessment: C2/5, explicit low-rank certification refinements are proved, but no decisive integral arithmetic step is replaced.

1 Scope, Status and the Source Argument

The assigned applicability entry is id 2, level L3, in the newly authorized L3 stage. L3 measures relevance and is not the contribution score. The subject is the October 3, 2026 manuscript [1], pinned at `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Its twelve build texts contain 7,502 lines and 361,048 bytes. This report does not claim a full-proof audit or a whole-corpus read. The report's theorems have proofs below; the source's unrestricted arithmetic statements are treated as imported when used.

Source/imported. For an elliptic curve $E/\mathbb{Q}$, put

$$s_q(E) = \operatorname{corank}_{\mathbb{Z}_q} \operatorname{Sel}_{q^\infty}(E/\mathbb{Q}), \quad a(E) = \operatorname{ord}_{s=1} L(E,s), \quad r(E) = \operatorname{rank} E(\mathbb{Q}).$$

The Selmer group is the full group, with local Kummer conditions at every place. The source claims, for every prime q and every $E/\mathbb{Q}$,

$$s_q(E) \in \{0,1\} \implies a(E) = r(E) = s_q(E), \quad \#\operatorname{Sha}(E/\mathbb{Q}) < \infty, \quad Q_E = \#\operatorname{Sha}(E/\mathbb{Q}), \tag{1.1}$$

where Q_E is defined in Section 2. The quantifier is not merely "Mordell–Weil rank at most one." It places no extra reduction, torsion, CM or residual-representation assumption on E. Nothing proved locally here establishes (1.1) at that generality.

1.1 Whole-source dependency map

All section and theorem locators were structurally scanned. The four fully read files are the main file, introduction, references and final center section. Other files were read only in bounded passages. The following map is a map of dependencies, not a list of verified proofs.

Build unit	Role in the source	Evidence boundary in this report
main; 01	Statement, conventions, one-sided versus unit comparisons	Full literal read; source theorem is reported, not independently certified.
02	Low-corank and two-primary companion inputs; auxiliary fields; Gross–Zagier and full-index identity	Inputs and exact real comparison read; auxiliary-field proof not inspected.
03	Finite cochain models, local orthogonality, line and square switches, Kummer volumes, character division	Determinant signs and Kummer/Haar passages read; most model and switching proofs imported.
04	Moving primes, residual concentration, obstruction matrices	Opening setup read; the residual concentration proof imported.
05	Single-curve determinant integrality; tame and spectral heights; central inequality $X_p(E) \geq 0$	Normalized coordinate and central calculation read; intervening integrality and height proofs imported.
06	Integral CM primitives, interpolation, vertical primitivity, horizontal divisibility	Opening setup read; construction of the analytic series and integral quotient imported.
07	Characteristic-zero theta tests, increasing p-level, uniform moment cuts and toric limits	Opening setup read; theta and increasing-level estimates imported.
08	Bounded content loss, cusp lifting, primitive Galois extension	Theta-implication statement read; the extraction proof imported.
09	Quotient association, simultaneous jumps of codimension two, removal of divisors	Final unit argument read; association and codimension lemmas imported.
10	Central exact volume, CM descent, all-prime assembly	Full literal read; algebraic bookkeeping checked, arithmetic prerequisites not reconstructed.
references	Classical and companion inputs	Full literal read; reading a bibliography is not reading its referenced proofs.

The source separates two tasks for an odd prime p: for non-CM E, the single-curve construction gives $X_p(E) \geq 0$; a suitable quadratic twist gives $X_p(E) + X_p(E^D) = 0$. Nonnegativity for both twists then forces equality. CM rank zero enters independently, followed by the pair identity in CM rank one. The two-primary companion supplies positivity, rationality and the valuation at two. This architecture explains why an accurate real number alone cannot replace the source's unit theorem.

1.2 Current primary literature, checked October 8, 2026

The publisher history [2] records an October 7 correction to an obsolete introductory citation in this manuscript. The current publisher README still identifies the October 3 title. The parent's authenticated batch identity record reports that all twelve assigned files agree with the captured October 8 publisher target; that authentication is parent evidence, not a fresh author download or an independent arithmetic review.

Jetchev–Skinner–Wan's primary abstract [3] treats semistable analytic-rank-one curves, good primes $p \geq 5$, and irreducible $E[p]$; it also states a qualified case at $p = 3$. Burungale–Flach [4] treats CM by the maximal order, nonzero L-value, and an abelian torsion-field extension over the CM field. Castella's revised primary abstract [5] gives rank-one CM results for $p > 3$ split in the CM field, with the stated abelian torsion-field condition. These are not silently enlarged into unrestricted all-prime inputs. Smith's primary abstract [6] gives the 50–50 distribution of full 2^∞-Selmer coranks in a quadratic-twist family; its advertised analytic Goldfeld consequence assumes BSD. It does not independently supply the source's unrestricted converse.

This is a bounded primary-record refresh. The four external research papers were read at abstract and version-metadata level, not fully audited. No conclusion about universal novelty or community acceptance follows. Searches used public titles only; private titles, proof text and this report were not sent to public search services.

2 Exact Normalizations and Arithmetic State

Let E be given by a global minimal Weierstrass equation, let N be its conductor, and let ω_E be its minimal Neron differential. Use the uncompleted Hasse–Weil L-function with every finite Euler factor. Write $t_E = \#E(\mathbb{Q})_{\mathrm{tors}}$ and $C_E = \prod_{\ell<\infty} c_\ell(E)$, where $c_\ell = [E(\mathbb{Q}_\ell) : E_0(\mathbb{Q}_\ell)]$. The source uses

$$\Omega_E = \int_{E(\mathbb{R})} |\omega_E|, \tag{2.1}$$

$$H(P) = \lim_{n\to\infty} 4^{-n} h_x([2^n]P), \qquad B(P,Q) = \tfrac{1}{2}\{H(P+Q) - H(P) - H(Q)\}. \tag{2.2}$$

Here $h_x(a/b) = \log \max(|a|, b)$ in lowest terms, $b > 0$, and $h_x(O) = 0$. In particular $B(P,P) = H(P)$. The canonical height of the origin divisor is $H/2$, as explicitly stated by the source. A software height or regulator must be converted to (2.2) before use. The period in (2.1) includes both real components if there are two; no additional real-component multiplier is appended.

On $V_E = (E(\mathbb{Q})/E(\mathbb{Q})_{\mathrm{tors}}) \otimes_{\mathbb{Z}} \mathbb{R}$, B is positive definite. The regulator is its determinant on a basis of the *full* free lattice, and is 1 in rank zero. For a finite-index point lattice L,

$$R_L = \det(B(P_i, P_j)) = I^2 \operatorname{Reg}_E, \qquad I = [E(\mathbb{Q})/E(\mathbb{Q})_{\mathrm{tors}} : L]. \tag{2.3}$$

Indeed an integral basis-change matrix A has $|\det A| = I$, and its Gram matrix is $A^T G A$. Positivity proves independence, not $I = 1$.

For common analytic and algebraic rank $r \in \{0, 1\}$, set

$$\ell_E = \frac{L^{(r)}(E,1)}{r!}, \qquad Q_E = \frac{\ell_E t_E^2}{\Omega_E \operatorname{Reg}_E C_E}, \qquad X_p(E) = v_p(Q_E) - v_p(S_E), \quad S_E = \#\operatorname{Sha}(E/\mathbb{Q}). \tag{2.4}$$

The definition of X_p requires finite S_E and nonzero rational Q_E. The source obtains positive rational Q_E through its companion input; this report does not infer rationality from a numerical enclosure. Without rank and finiteness inputs, the analytic expressions below still make sense, but Q_E need not have this BSD interpretation.

2.1 Private proof route and restricted transfer

The actual private file `TR-item-004-astra.tex` [7] was read at its Mellin/log-moment proof, coefficient and spatial tails, panel proof, combined enclosure, rank decision, regulator proof, saturation gate, discrete-isolation proof and revised normalization/tail sections. These are inherited tools, not new results of this report. The projection proof in the actual private TR250 [8] was also read. On paired complex samples (v_+, v_-), the map

$$P_w(v_+, v_-) = \tfrac{1}{2}(v_+ + wv_-,\ v_- + wv_+), \qquad w \in \{\pm 1\},$$

is an orthogonal projection in the equal-weight Euclidean metric. If the true pair has that parity, projection cannot increase its error. This is an elementary instance of the private constraint principle, not a new arithmetic theorem. A changed sampling metric requires its own orthogonal projection. Signed newform coefficients are not projected onto a positive cone.

3 Low-Rank Mellin Reduction with No Derivative Differencing

Put $\alpha = 2\pi/\sqrt{N}$, and write the normalized rational newform as $f_E(\tau) = \sum_{n\ge 1} a_n e^{2\pi i n\tau}$, $a_1 = 1$. Then

$$g(x) = f_E(ix/\sqrt{N}) = \sum_{n\ge 1} a_n e^{-\alpha n x}, \qquad \Lambda(E,s) = N^{s/2}(2\pi)^{-s}\Gamma(s)L(E,s).$$

The functional equation is $\Lambda(E,s) = w\Lambda(E,2-s)$ and $g(1/x) = wx^2 g(x)$. Cusp decay gives

$$\Phi(z) := \Lambda(E, 1+z) = \int_1^\infty g(x)(x^z + wx^{-z})\,\mathrm{d}x. \tag{3.1}$$

This is the inherited paired Ram log-atom representation. Its domain is contained in the weighted function space

$$\bigcap_{k\ge 0} L^1([1,\infty), (\log x)^k\,\mathrm{d}x),$$

with the stronger cusp decay needed for locally uniform complex differentiation. Our g has that decay. It is not a positive measure.

For every integer $k \ge 0$,

$$\Phi^{(k)}(0) = (1 + w(-1)^k)\int_1^\infty g(x)(\log x)^k\,\mathrm{d}x. \tag{3.2}$$

Dominated differentiation on each compact set in z proves this. The standard coefficient envelope $|a_n| \le d(n)\sqrt{n} \le 2n$, including the usual bad-prime coefficients, makes all subsequent termwise integrations absolutely convergent. Modularity and the coefficient bound are classical inputs, not local discoveries.

Proposition 3.1 (Two scalar leading-term kernels)**.** *For every such E, define*

$$K_0(\beta) = \frac{e^{-\beta}}{\beta}, \qquad K_1(\beta) = \int_1^\infty e^{-\beta x}\log x\,\mathrm{d}x = \frac{E_1(\beta)}{\beta}, \quad \beta > 0,$$

where $E_1(\beta) = \int_\beta^\infty e^{-u}\,\mathrm{d}u/u$. If $w = 1$, then $L(E,1) = 2\alpha\sum_{n\ge 1} a_n K_0(\alpha n)$. If $w = -1$, then $L'(E,1) = 2\alpha\sum_{n\ge 1} a_n K_1(\alpha n)$. When these quantities are nonzero, the analytic rank is respectively zero and one.

Proof. For $k = 0$ integrate the exponential; for $k = 1$, integration by parts gives $K_1 = \beta^{-1}\int_1^\infty e^{-\beta x}\,\mathrm{d}x/x$ and then substitute $u = \beta x$. Since the completion factor at one is $1/\alpha$, $L(1) = \alpha\Phi(0)$. For $w = -1$, $\Phi(0) = L(1) = 0$ exactly, so the product rule gives $L'(1) = \alpha\Phi'(0)$. Absolute convergence justifies the sums. Nonvanishing and the exact odd-parity zero give the ranks. □

For $w = 1$, the identity for $L'(1)$ would require the lower product term and is not the rank-one formula above. In arbitrary order the correct conversion is

$$L^{(k)}(1) = \sum_{j=0}^{k} \binom{k}{j} (A^{-1})^{(k-j)}(1)\Phi^{(j)}(0), \quad A(s) = N^{s/2}(2\pi)^{-s}\Gamma(s).$$

An interval containing zero is never an exact vanishing certificate. The exact rank assumption from a Selmer theorem must not be replaced by an empirical smallness test.

4 A Sharper Rank-One Certificate

The private general-order method already suggests exact integration of exponentials. Low rank permits a more specific positive-kernel construction with explicit rational bounds. Only the kernel is positive; the newform and its signed sum need not be.

Theorem 4.1 (Rational kernel bracket)**.** *For every $\beta > 0$ let*

$$F(\beta) = \int_0^\infty \frac{e^{-t}}{\beta + t}\,\mathrm{d}t, \qquad K_1(\beta) = \frac{e^{-\beta}}{\beta}F(\beta).$$

Then

$$\frac{1}{\beta+1} \le F(\beta) \le \frac{\beta+1}{\beta(\beta+2)} < \frac{1}{\beta}. \tag{4.1}$$

The width of the induced K_1 bracket is exactly

$$\frac{e^{-\beta}}{\beta^2(\beta+1)(\beta+2)}. \tag{4.2}$$

Its midpoint has absolute error at most half this width. Relative to the positive lower bound for K_1, that error is at most $1/[2\beta(\beta+2)]$.

Proof. The change $u = \beta + t$ in E_1 gives the representation. Jensen's inequality for the probability density e^{-t}, of mean one, gives $F \ge 1/(\beta+1)$. Also

$$F = \frac{1}{\beta} - \frac{1}{\beta}\int_0^\infty \frac{te^{-t}}{\beta+t}\,\mathrm{d}t.$$

The density te^{-t} has total mass one and mean two. Jensen gives the last integral at least $1/(\beta+2)$, proving the upper bound. Subtracting endpoints gives $1/[\beta(\beta+1)(\beta+2)]$; the prefactor yields (4.2) and the relative statement. □

This bracket is inexpensive but is not arbitrarily accurate at a fixed small β. Increasing the number of Fourier coefficients alone does not shrink the bracket of the first coefficient. The next theorem supplies the necessary refinement instead of hiding that error floor.

Theorem 4.2 (Refinable weighted rational panels)**.** *Fix $\beta > 0$, $T > 0$, a finite partition of $[0, T]$ into panels $[a, b]$, and an integer $m \geq 0$. For a panel let $c = (a + b)/2$, $h = (b - a)/2$ and*

$$S_{m,[a,b]}(\beta) = \sum_{j=0}^{m} \frac{(-1)^j}{(\beta + c)^{j+1}} \int_a^b e^{-t}(t - c)^j \, \mathrm{d}t.$$

Then its exact contribution to F differs from $S_{m,[a,b]}$ by at most

$$R_{m,[a,b]} = \frac{h^{m+1}(e^{-a} - e^{-b})}{(\beta + c)^{m+1}(\beta + a)}. \tag{4.3}$$

The omitted tail satisfies

$$\frac{e^{-T}}{\beta + T + 1} \leq \int_T^\infty \frac{e^{-t}}{\beta + t} \, \mathrm{d}t \leq \frac{e^{-T}}{\beta + T}. \tag{4.4}$$

Summed panel intervals plus this tail, intersected with (4.1), enclose F. With rigorous arithmetic enclosures, they can be refined to any prescribed positive absolute error.

Proof. The exact finite geometric identity is

$$\frac{1}{\beta + t} = \sum_{j=0}^{m} \frac{(-1)^j (t - c)^j}{(\beta + c)^{j+1}} + \frac{(-1)^{m+1}(t - c)^{m+1}}{(\beta + c)^{m+1}(\beta + t)}.$$

Bound the last denominator below by $\beta + a$, use $|t - c| \leq h$, and integrate the positive weight. Shifting $t = T + u$ proves the tail upper bound directly and its lower bound by the first Jensen bound. The panel moments are finite endpoint exponential-polynomial expressions, obtained by integration by parts; their odd powers must be retained because e^{-t} is not symmetric about c. For a fixed finite partition, $h/(\beta + c) < 1$ on every panel, so increasing m sends every remainder to zero. Increasing T controls the tail and increasing arithmetic precision controls rounding. This proves existence of a terminating refinement for a fixed β and positive tolerance, not a uniform conductor-complexity bound. □

Corollary 4.3 (Certified Fourier tails for the uncompleted term)**.** *Let $M \geq 1$, $n_0 = M + 1$, and $\beta_0 = \alpha n_0$. With the appropriate root number, the omitted tail in Proposition 3.1 obeys*

$$B_0(M) = \frac{4e^{-\beta_0}}{1 - e^{-\alpha}}, \tag{4.5}$$

$$B_1(M) = 4\frac{\beta_0 + 1}{\beta_0(\beta_0 + 2)} \frac{e^{-\beta_0}}{1 - e^{-\alpha}}. \tag{4.6}$$

The rank-one bound is the inherited order-one log-atom majorant, converted to L', multiplied by $(\beta_0 + 1)/(\beta_0 + 2) < 1$.

Proof. For rank zero, $2\alpha|a_n|K_0(\alpha n) \leq 4e^{-\alpha n}$. For rank one, Theorem 4.1 gives $2\alpha|a_n|K_1(\alpha n) \leq 4e^{-\alpha n}U(\alpha n)$, where $U(\beta) = (\beta + 1)/[\beta(\beta + 2)] = \frac{1}{2}(1/\beta + 1/(\beta + 2))$ is decreasing. Bound it by $U(\beta_0)$ and sum the geometric series. The private order-one bound is $4e^{-\beta_0}/[\beta_0(1 - e^{-\alpha})]$ after multiplication by α. Their ratio is as stated. □

For a signed finite coefficient list, enclose each $K_r(\alpha n)$ and multiply its interval by the signed exact a_n, swapping endpoints when $a_n < 0$. Sum, multiply by 2α, and add $[-B_r, B_r]$ and the rounding enclosure. Exact algebra with α is a specification; an implementation must enclose π, $\sqrt{N}$, exponentials and every operation outwardly. Interval evaluation on a certified interval for α is sufficient. No real curve computation is claimed by the finite smoke tests accompanying this report.

4.1 What improves, and what does not

Rank zero has an exact elementary coefficient kernel. Rank one has a kernel interval with an explicit error and no need to numerically differentiate L. The private method's spatial cutoff and derivative remainder are replaced by a weighted rational-panel remainder and a simple positive tail. This is a change of certifying representation, not elimination of all approximation. The factor in (4.6) is strictly smaller but approaches one for large β_0; it is a modest gain, not a complexity breakthrough. Cancellation among signed terms and large conductors remain costly.

5 Integral Defects, Indices and Unit Inputs

The source's strict determinant uses T_pE, not just V_pE. For its inverse determinant convention, a generically acyclic complex in degrees $1, 2$ has $d(C, 1) = -v_p(\det d^1)$. Torsion in degree i contributes $(-1)^{i+1}$ times its length. These signs are imported conventions and retained below.

At an odd prime p, the source chooses $K = \mathbb{Q}(\sqrt{D})$, with p and the level primes split and the required auxiliary hypotheses. Its point P_K is the Hilbert-class trace *without division by the class number*. For $\phi : X_0(N) \to E$,

$$\phi^*\omega_E = c_E f \frac{dq}{q}.$$

The exact rational Manin factor c_E is not assumed a unit. Define

$$n_P = v_p([E(K)/E(K)_{\mathrm{tors}} : \mathbb{Z}\overline{P}_K]), \quad \tau_g = v_p(\#E(K)_{\mathrm{tors}}), \quad s_K = v_p(\#\,\mathrm{Sha}(E/K)).$$

The source's real comparison gives

$$\frac{Q_E Q_{E^D}}{S_E S_{E^D}} = \frac{[E(K) : \mathbb{Z}P_K]^2}{c_E^2 C_E^2 \#\,\mathrm{Sha}(E/K)}, \qquad [E(K) : \mathbb{Z}P_K] = I_K \# E(K)_{\mathrm{tors}}. \tag{5.1}$$

Here I_K is the index in the free rank-one group. The regulator over K uses the relative diagonal $2H_{\mathrm{abs}}$; this field-degree factor is already built into the real comparison, not added a second time.

Proposition 5.1 (Defect-preserving central comparison)**.** *Assume the source's exact central determinant and interpolation formulas and its real comparison, but do not assume its final unit theorem. Write $h = v_p(\log_{\omega_E} P_K)$, $\log_{\omega_E} E(\mathbb{Q}_p) = p^l\mathbb{Z}_p$, $\tau_q = v_p(\#E(\mathbb{Q}_q)[p^\infty])$, $e_q = v_p(L_q(E, 1)^{-1})$, and $c = v_p(c_E)$. Let $B_w(0)$ and $L_w(0)$ be the nonzero central values in those formulas. Then*

$$\delta_p := v_p(B_w(0)) - v_p(L_w(0)) = X_p(E) + X_p(E^D) = 2n_P + 2\tau_g - s_K - 2c - 2\sum_q v_p(c_q(E)). \tag{5.2}$$

Thus the source's integral-unit comparison implies $\delta_p = 0$; mere integrality of a nonzero central quotient implies only $\delta_p \geq 0$.

Proof. The two assumed source formulas are

$$-v_p(L_w(0)) = 2n_P + 2\tau_g - s_K - 2(h - l + \tau_p) - 2\sum_{q\in S^\circ} \tau_q,$$

$$v_p(B_w(0)) = -2c + 2h + 2e_p + 2\sum_{q\in S^\circ} e_q.$$

Add them. The $2h$ terms cancel. The exact Haar identities $\tau_p - l - e_p = v_p(c_p(E))$ and $\tau_q - e_q = v_p(c_q(E))$ identify the remaining local terms. All bad primes are in the support and omitted good primes have component factor one. This proves the last expression in (5.2); taking the valuation of (5.1) proves the middle expression. A unit specializes to a unit under the integral augmentation $\mathbb{Z}_p[[t, \mathbf{u}]] \to \mathbb{Z}_p$. □

This is a checked cancellation conditional on the source arithmetic formulas, not a reconstruction of those formulas. Deleting τ_g shifts the expression by $-2\tau_g$; deleting the local τ_p shifts it by $2\tau_p$. Such errors are invisible after rationalizing finite modules. Likewise an equality up to a rational scalar is not an integral unit equality. Real error bars do not bound δ_p.

Lemma 5.2 (Parametrization and lattice covariance)**.** *Replacing ϕ by $[k]\circ\phi$, $k \in \mathbb{Z}\backslash\{0\}$, replaces P_K by kP_K and c_E by kc_E. Consequently the right side of (5.1), and the expression (5.2), are unchanged. Replacing a rank-one rational point P by kP changes the subgroup quotient $q_P := \ell_E t_E^2/(\Omega_E H(P) C_E)$ to q_P/k^2 and its free index I to $|k|I$, leaving $I^2 q_P = Q_E$ unchanged.*

Proof. Pullback of the invariant differential by $[k]$ is multiplication by k. The free index is multiplied by $|k|$, and $H(kP) = k^2 H(P)$. The indicated squares cancel. In valuations, both n_P and c increase by $v_p(k)$. This proves both assertions. □

Changing P_K alone while keeping c_E and its original trace normalization fixed is not this covariance and is not licensed by it.

6 A Conditional Arithmetic Interval Bridge

For rank one, assume that rank is proved and P is nontorsion. Suppose $\ell_E \in [\ell_-, \ell_+]$, $\Omega_E \in [\Omega_-, \Omega_+]$ and $H(P) \in [h_-, h_+]$, with every lower endpoint positive. Exact torsion and component data give the rigorous subgroup quotient interval

$$q_P \in J = \left[\frac{\ell_- t_E^2}{\Omega_+ h_+ C_E}, \frac{\ell_+ t_E^2}{\Omega_- h_- C_E} \right]. \tag{6.1}$$

For rank zero omit the height factor and set $I = 1$. Correlation-aware interval evaluation may narrow J but is not needed for containment. Logarithmic relative budgets may be added only after positivity has been certified.

Lemma 6.1 (A finite saturation boundary)**.** *In rank one, suppose an independent theorem supplies $h_* > 0$ such that $H(Q) \geq h_*$ for every nontorsion $Q \in E(\mathbb{Q})$. If $H(P) \leq h_+$, then the free index satisfies*

$$I \leq B := \left\lfloor \sqrt{h_+/h_*} \right\rfloor.$$

If the subgroup generated by P is proved p-saturated in the full free group for every prime $p \leq B$, then $I = 1$.

Proof. Write $P = IP_0$ modulo torsion for a primitive generator P_0. Then $I^2 h_* \leq I^2 H(P_0) = H(P) \leq h_+$. If $I > 1$, some prime $p \mid I$ satisfies $p \leq I \leq B$ and obstructs p-saturation. □

The independent lower height bound and the prime-saturation proofs are additional arithmetic data, not consequences of a positive height interval. A point search does not furnish either certificate.

Theorem 6.2 (Odd-integer discrepancy test with bounded index)**.** *Assume common rank zero or one, finite* Sha*, and an independently proved exact order $S = S_E$. Assume also $Q_E/S \in \mathbb{Q}_{>0}$ and*

$$v_2(Q_E/S) = 0, \qquad v_p(Q_E/S) \geq 0 \quad \text{for every odd prime } p. \tag{6.2}$$

Let $1 \leq I \leq B$ be a proved bound on the index of the point lattice used in q_P, and let $q_P/S \in [a, b]$ be rigorous with $a > 0$. If $B^2 b < 3$, then $Q_E = S$. More generally, if an independent argument gives zero valuation at every prime in a finite set $\mathcal{P}$ containing two, replace three by the smallest prime p_* not in $\mathcal{P}$.

Proof. Every prime valuation of the positive rational $D = Q_E/S$ is nonnegative, so its reduced denominator is one. Its valuation at two vanishes, so it is a positive odd integer. By (2.3), $D = I^2 q_P/S \le B^2 b < 3$, hence $D = 1$. In the general case, D is a positive integer coprime to every prime in $\mathcal{P}$. Any integer $D > 1$ has a prime divisor outside $\mathcal{P}$, so $D \ge p_*$. The same strict upper bound proves the claim. □

Source/imported. For a non-CM low-rank curve, the source's single-curve inequality and two-primary input have exactly the shape (6.2). **Local deduction.** The theorem gives an alternative final step *for an individual curve with the stated extra certificates*. It does not prove those all-prime inequalities or independently compute S. It does not apply the non-CM single-curve result to CM curves without another input. Compared with the source's pair/unit route, it trades a uniform arithmetic argument for explicit curve-specific data. The exact order of Sha may be the hardest unavailable datum.

If the possible indices form a proved finite set $\mathcal{I}$, retain the union $\bigcup_{j\in\mathcal{I}} j^2[a,b]$ and intersect it with the positive odd integers. If this intersection is $\{1\}$, the same proof works. Taking a convex hull too early can discard useful gaps. Empty intersection means inconsistent certificates, not a BSD proof. For example, $[a,b] = [1/10, 3/25]$ and $\mathcal{I} = \{1,2,3\}$ leave only the odd integer one in the union; this is a manufactured certificate example, not an elliptic-curve calculation. Under the assumptions it forces $I = 3$ and $D = 1$.

6.1 Why weaker arithmetic promises do not suffice

Positive rationality and the correct two-adic valuation alone do not give a discrete real set: $(2m+1)/(2m-1)$ has two-adic valuation zero and tends to one from above. The all-prime nonnegativity in (6.2) is doing essential work. If instead an independent theorem supplies $dQ_E \in \mathbb{Z}$ for a known integer $d > 0$, an interval with a singleton intersection with $d^{-1}\mathbb{Z}$ identifies Q_E; width less than $1/d$ only guarantees at most one candidate, not its existence. If only reduced denominators at most D_0 are known, distinct candidates differ by at least $1/D_0^2$. Neither denominator promise follows from rationality alone. Identifying Q_E still must be separated from proving that it equals the independently determined order of Sha.

7 Verification and Practical Use

The accompanying bounded checks exercise the rational bracket identity, weighted-panel geometric remainder, signed coefficient tails, height and index covariance, odd-integer isolation and central valuation cancellation. Rational identities use exact fractions. Numerical kernel comparisons use high-precision arithmetic as smoke tests, not as outward-rounded theorem certificates. No tested coefficient list is represented as an elliptic-curve newform unless separately proved; no such real-curve computation is claimed here.

A proof-producing implementation should separate the following records.

1. Minimal equation, conductor, root number, coefficient list and exact torsion/component computations, each with its arithmetic source.

2. A rigorous interval for the rank-appropriate leading term, including signed kernel intervals, Fourier tail and rounding error.

3. The total real period and the exact height convention, followed by intervals for the displayed point lattice, not an assumed full basis.

4. A saturated-basis certificate or a proved finite index set/bound.

5. Arithmetic assumptions required for the selected decision rule, including rationality, all-prime valuations, and any exact Sha order.

6. The decision: analytic rank certified, subgroup quotient enclosed, conditional equality obtained, or unresolved. These are different outputs.

An implementation must fail without making an equality claim whenever one of the required records is absent. A failed positivity test does not mean a negative leading term; it can mean insufficient precision.

7.1 Advantages, costs and applicability

Advantages. The leading-term kernels remove differentiation noise, enforce parity exactly, and support finite local error budgets. The sharper tail uses the actual low-rank kernel. The determinant-defect identity preserves finite torsion and detects convention mistakes. The interval bridge can avoid computing an exact point index when a sufficient bound is already available.

Costs and limitations. The Fourier length grows when $\alpha = 2\pi/\sqrt{N}$ is small; near cancellation increases the precision burden. The crude rational bracket alone has a nonzero error floor. Period, local reduction, height and saturation algorithms are outside the analytic certificate. The arithmetic bridge has strong global hypotheses and is not a replacement for the general BSD proof. No runtime comparison with modular symbols, approximate functional equations or established L-function packages was performed. For rank-zero exact rational values, modular symbols remain a natural independent route; for rank one the kernel certificate is a check on the analytic side, not an Euler system.

The source's p-adic theta moment estimates and the present real log-moment estimates are different objects in different topologies. Similarity of their names does not produce an integral comparison functor. Neither the private Ram representation nor a change of coordinates proves an integral lattice is saturated.

8 Contribution and Further Research

Realized local results. Theorems 4.1 and 4.2 and Corollary 4.3 provide explicit low-rank analytic refinements of the inspected private route. Proposition 5.1 retains the missing unit defect instead of assuming it zero. Lemma 5.2 records the simultaneous point and differential scaling. Theorem 6.2 gives a source-specific conditional interval bridge with an index bound and a threshold of three, exploiting the exact two-primary input. Their ingredients are classical analysis, lattice algebra and elementary rational arithmetic; no historical priority is claimed.

Assessment: C2/5. These are modest but proved improvements to certification and bookkeeping. The contribution is not raised because the source theorem is important or because the report is long. Low-corank converse, auxiliary-field availability, integral determinant primitivity, theta extraction, quotient-unit removal and exact arithmetic orders remain imported. No full BSD implication follows from the finite tests, and no contribution credit is assigned to the parent's checks.

Useful next steps are a directed-rounding implementation of the weighted panels with independently generated newform coefficients; low-rank benchmark comparisons against certified modular-symbol or approximate functional-equation computations; and integration with independently proved saturation and height-lower-bound records. One should compare total bit complexity, not just the tail constant. A separate arithmetic project could test which curve classes admit

independently known all-prime one-sided inequalities and Sha orders for the interval bridge. None of these proposed computations or arithmetic extensions is counted as completed work.

A Literal Reading and Evidence Boundaries

All scientific inputs were hashed before the literal reads and are checked again at finalization. Equality of endpoint hashes is not continuous change monitoring. Inclusive line intervals below record actual displayed, untruncated source text, not the results of a search for a label. Headings and labels were additionally scanned across all twelve build files. They do not convert unread lines into proof reads.

Source file	Literal line intervals	Lines
main.tex	1–82	82
references.tex	1–115; 116–230	230
01-introduction.tex	1–169; 170–281	281
02-classical.tex	1–98; 196–326	229
03-models.tex	296–339; 449–538	134
04-vertical.tex	1–52	52
05-single.tex	361–427; 1274–1379	173
06-pair.tex	1–52	52
07-theta.tex	1–52	52
08-extraction.tex	1–52	52
09-unit.tex	391–449	59
10-center.tex	1–167; 168–271	271
Total source literal read	Four complete files; eight partial	1667
Private item 004	198–307; 395–470; 471–634; 676–770; 867–1052	631
Private TR250	668–724	57

The source has 7,502 lines; 5,835 are not claimed literally read. The administrative README, M719 evidence conventions, contribution policy and existing compile/freeze helpers were read. The shared catalogue was inspected only at row 2. Parent source authentication and the parent's seven finite convention checks are external batch evidence and are not included in the author test count. The report's audit directory records exact paths, SHA-256 values, build diagnostics, finite tests, all-page visual review and the final freeze. The source authors' original files and the private corpus were not changed; no shared queue, catalogue or progress file was edited.

References

[1] OpenAI, *Exact Birch–Swinnerton-Dyer Formula from Low Selmer Corank*, October 3, 2026. Pinned build source in this project's source cache. Pinned publisher manuscript.

[2] OpenAI, *Math release history*, October 7, 2026 entry, checked October 8, 2026. https://raw.githubusercontent.com/openai/math/main/history.md.

[3] D. Jetchev, C. Skinner and X. Wan, *The Birch and Swinnerton-Dyer Formula for Elliptic Curves of Analytic Rank One*, primary abstract and metadata. https://arxiv.org/abs/1512.06894.

[4] A. Burungale and M. Flach, *The conjecture of Birch and Swinnerton-Dyer for certain elliptic curves with complex multiplication*, version 3, September 4, 2023; primary abstract. https://arxiv.org/abs/2206.09874v3.

[5] F. Castella, *Tamagawa number conjecture for CM modular forms and Rankin–Selberg convolutions*, version 3, September 30, 2025; primary abstract and metadata. https://arxiv.org/abs/2407.11891.

[6] A. Smith, *The Birch and Swinnerton-Dyer conjecture implies Goldfeld's conjecture*, March 22, 2025; primary abstract. https://arxiv.org/abs/2503.17619.

[7] M. Subbarao, confidential local proof source `TR-item-004-astra.tex`. Selected original proof text only; literal scopes and hash are recorded in the author read ledger. No public upload or whole-corpus-read claim.

[8] M. Subbarao, confidential local proof source `TR250-Compatible-Variational-Ram-Panels.tex`, lines 668–724. Metric projection proof and its limitations; not a full-file read.